

Modelling Competition in Telecommunications

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Abstract—This paper presents the use of a Markov decision process (MDP) model to predict changes in market share in the telecommunications industry. In this model, at each decision epoch, or time step, each decision making provider has a set of allowable actions that can be applied with varying levels of intensity. Providers make decisions and apply their actions independently. These combined actions mean that at each decision epoch we use a different transition probability matrix. Consequently, rewards can change depending on the decisions made at each step. The MDP formulation and two numerical examples are included, to illustrate that MDPs are appropriate for modelling competition.

Index Terms—Telecommunications, market share analysis, Markov chain, Markov decision process, decision making.

I. INTRODUCTION

The telecommunication industry is usually made up of a few major players, plus many minor players in a market. The major players are usually incumbent telephone companies or operators that own and have control over physical telecommunications infrastructure. They can provide telecommunication services directly to subscribers, and they can wholesale their services to downstream providers. This paper explores mobile market shares captured by providers who sell directly to retail subscribers, regardless of whether or not they own infrastructure. The mobile industry is fast moving and competition is fierce. It is desirable for providers to have a model to predict market share, determine expected earnings, and evaluate strategies or counter actions in response to their competitors.

The underlying mathematical model that we will use is a finite-horizon, discrete time Markov decision process (MDP). The general theory for MDPs, notation, and terminology are set out in [1]. Put simply, a MDP is a discrete time stochastic control process. At each decision epoch, a set of actions is allowed for each state. The decision maker chooses an action from a set of allowable actions, and the process moves randomly from one state to a new state, and gives the decision maker a reward accordingly. MDPs can be considered as

Markov chains with added actions and rewards. When there is no action applied, an MDP becomes a Markov chain (MC) with a constant transition probability matrix (transition matrix). When there is action, the transition matrix changes, the reward changes, and the MC becomes time-inhomogeneous. Most MDP applications aim to find the optimal policy, or a sequence of actions that gives the best reward over a period of time.

Fig. 1 depicts a 2-state MDP. When no action is taken at decision epoch t , the process is time-homogeneous, and the transition probabilities are constant. When actions are applied, the process dynamics change and consequently the transition probabilities will change.

Markov chains have often been used as models in market research. An early example in [2] presented a Markovian analysis of newspaper subscriptions. Similarly, it was proposed in [3], that mobile subscriptions offer a more contemporary example to introduce students to Markov chains. References [4] and [5] have used Markov chains to model market shares of mobile operators. In these works, the probabilities of switching providers were represented in transition probability matrices, and Markov chains were used to forecast future market shares. Reference [6] included a diffusion growth model in the Markov formulation that forecasts changes in market share according to phases in the product life cycles.

By contrast, the present work uses Markov Decision Processes to model market shares of mobile operators as customers move between providers. We seek to use MDP framework to model market share and include actions, decisions and policies, and rewards. Service providers may take actions at certain levels of intensity at any decision epoch. Their actions cause changes to the transition matrix, and the corresponding rewards to the decision maker, at that epoch. The paper presents a model and numerical examples.

II. NOTATION

The notation used in the MDP model is presented below.

A. Notation for Spaces

- T : Decision epoch space, $T = \{1, 2, \dots, N\}$, consists of all decision epochs.
- S : State space, $S = \{s_1, s_2, \dots, M\}$, consists of all mobile providers.
- A_s : Action space for provider s , $A_s = \{a_{s,1}, a_{s,2}, \dots, a_{s,K_s}\}$, consists of all allowable actions for provider s .

B. Notation for Vectors and Vector Elements

- X_t : Distribution vector of system states at decision epoch t ; X_1 is the initial distribution vector. When a policy π is followed, we write $X_t^\pi(s)$.

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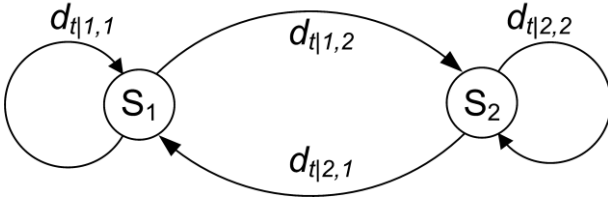


Fig. 1. Graphical representation of a 2-state MDP.

- $x_t(s)$: An element of the vector X_t , the market share of provider s at decision epoch t . This is the probability of being a customer of mobile provider s . When policy π is followed, we write $x_t^\pi(s)$.
- Y_t : User (subscriber) number vector at decision epoch t , Y_1 is the initial user number vector. When a policy π is followed, we write $Y_t^\pi(s)$.
- $y_t(s)$: An element of the user number vector Y_t , the number of users of provider s at decision epoch t . When a policy π is followed, we write $y_t^\pi(s)$.
- $L_t(s)$: Action intensity vector of provider s at decision epoch t . When policy π is followed we write $L_t^\pi(s)$.
- $l_t(s, k)$: Intensity of the k^{th} action, $a_{s,k}$, taken by provider s , at decision epoch t , $1 \leq k \leq K_s$. When policy π is followed, we write $l_t^\pi(s, k)$.
- V_t : Value vector at decision epoch t . Value can be revenue, profit, loss, cost, or other measures. When policy π is followed, we write V_t^π .
- $v_t(s)$: Value contributed to mobile provider s at decision epoch t . When policy π is followed, we write $v_t^\pi(s)$.
- W_t : Total value vector, the total reward accumulated over a time period from the beginning to decision epoch t . When policy π is followed, we write W_t^π .
- $w_t(s)$: Total reward contributed to mobile provider s , accumulated from the beginning to decision epoch t . When policy π is followed, we write $w_t^\pi(s)$.

C. Notation for Matrices and Matrix Elements

- P : Transition probability matrix under the Business As Usual (BAU) condition.
- $p_{i,j}$: Element of P that is the probability to churn users from mobile provider i to j under the BAU conditions.
- $\Delta(s, k)$: Action delta matrix, incremental probability added to P if mobile provider s takes the k^{th} action, $a_{s,k}$.
- $\delta_{i,j}(s, k)$: Incremental probability to churn users from provider i to j as a result of mobile provider s taking the k^{th} action, $a_{s,k}$.
- D_t : Applied action matrix, the transition matrix combining actions taken by all providers at decision epoch t . When a policy π is followed, we write D_t^π .

$d_{t|i,j}$: Transitional probability to churn from provider i to provider j at decision epoch t . When a policy π is followed, we write $d_{t|i,j}^\pi$.

R_j : Reward matrix for mobile service provider j .

$r_{i,j}$: Element of R_j , reward for provider j when a subscriber churns from provider i to j .

III. MODEL FORMULATION

Decisions can be made by one or more providers at discrete points of time referred to as decision epochs. Let T denote the set of decision epochs,

$$t \in T, T = \{1, 2, \dots, N\}, N \leq \infty.$$

At each decision epoch, a user may obtain mobile service from any one of the providers. Let S denote the set of providers,

$$s \in S, S = \{s_1, s_2, \dots, M\} = \{1, 2, \dots, M\}, M < \infty.$$

In the beginning, at decision epoch $t = 1$, we use the distribution vector X_1 to represent the initial market shares of providers,

$$X_1 = [x_1(s_1) \quad x_1(s_2) \quad x_1(s_3) \quad \dots \quad x_1(s_N)],$$

and vector Y_1 to represent the numbers of users of providers,

$$Y_1 = uX_1 = [y_1(s_1) \quad y_1(s_2) \quad y_1(s_3) \quad \dots \quad y_1(s_N)],$$

where u is the user population or total number of users of all providers.

Each provider s is a decision maker, and has its own set of allowable actions. Providers make decisions independently. At each decision epoch, each provider chooses to take or not to take any action, and to set intensities of its actions. As an example, the action “to reduce price” can be applied with “modest” or “high” intensity. In our model, the intensity is expressed in a scale between 0 and 1.

Let A_s be the action space for a mobile provider s ,

$$a_s \in A_s, A_s = \{a_{s,1}, a_{s,2}, \dots, a_{s,K_s}\}.$$

For example, for provider s_2 ,

$$a_{s_2} \in A_{s_2}, A_{s_2} = \{a_{s_2,1}, a_{s_2,2}, \dots, a_{s_2,K_{s_2}}\} = \{1, 2, \dots, K_{s_2}\}.$$

For each action, there is an associated incremental probability matrix, $\Delta(s, k)$, that can be scaled and added to the BAU transition probability matrix, P , as a result of mobile provider s taking the k^{th} action, $a_{s,k}$, to a certain level of intensity. We will call $\Delta(s, k)$ the delta matrix for simplicity. For example, provider s_2 has a total of K_{s_2} delta matrices:

$$\Delta(s_2, 1), \Delta(s_2, 2), \dots, \Delta(s_2, K_{s_2}).$$

The entries of the delta matrix, $\Delta(s, k)$, are given as $\delta_{i,j}(s, k)$, where

$$\sum_{j=1}^M \delta_{i,j}(s, k) = 0, s \in S, k \in A_s, i = 1, 2, \dots, M. \quad (1)$$

A decision maker sets the $\delta_{i,j}(s, k)$ values subjectively. For each provider, the total number of users at decision epoch t is always equal to the total of increase in users who choose to stay, and decrease in those who choose to leave at decision epoch $t + 1$. Therefore, the sum of all entries in a row of each $\Delta(s, k)$ must be zero.

To model the intensity of an action, a scaling factor, $l_t(s, k)$, ranging from 0 to 1 is used. At decision epoch t , the intensities of all allowable actions for provider s are put in vector form,

$$L_t(s) = [l_t(s, 1) \quad l_t(s, 2) \quad \dots \quad l_t(s, K_s)], 0 \leq l_t(s, k) \leq 1.$$

For example, at decision epoch $t = 3$, if provider s_1 takes action 2, $a_{s_1,2}$, at full intensity, and provider s_2 takes action 3, $a_{s_2,3}$, at half intensity, the action intensity vectors for providers s_1 and s_2 would be:

$$\begin{aligned} L_3(s_1) &= [l_3(s_1, 1) \quad l_3(s_1, 2) \quad l_3(s_1, 3) \quad \dots \quad l_3(s_1, K_{s_1})] \\ &= [0 \quad 1 \quad 0 \quad 0 \quad \dots \quad 0], \\ L_3(s_2) &= [l_3(s_2, 1) \quad l_3(s_2, 2) \quad l_3(s_2, 3) \quad \dots \quad l_3(s_2, K_{s_2})] \\ &= [0 \quad 0 \quad 0.5 \quad 0 \quad \dots \quad 0]. \end{aligned}$$

The transition probability matrix combining all actions taken by all providers at decision epoch t is given as:

$$D_t = P + \sum_{s=1}^M \sum_{k=1}^{K_s} l_t(s, k) \Delta(s, k) \quad (2)$$

To explain how we can combine the BAU transition probability matrix P and an incremental probability matrix $\Delta(s, k)$ with an applied intensity of $l_t(s, k)$, we use a simple example in which there is only one action, a_{s^*,k^*} , taken by provider $s = s^*$, at half intensity, i.e. $l_{t^*}(s^*, k^*) = 0.5$, at decision epoch $t = t^*$. The P and $\Delta(s^*, k^*)$ matrices are:

$$\begin{aligned} P &= \begin{bmatrix} p_{1,1} & \dots & p_{1,M} \\ \vdots & \ddots & \vdots \\ p_{M,1} & \dots & p_{M,M} \end{bmatrix}, \text{ and} \\ \Delta(s^*, k^*) &= \begin{bmatrix} \delta_{1,1}(s^*, k^*) & \dots & \delta_{1,M}(s^*, k^*) \\ \vdots & \ddots & \vdots \\ \delta_{M,1}(s^*, k^*) & \dots & \delta_{M,M}(s^*, k^*) \end{bmatrix}. \end{aligned}$$

Using equation (2), the transition probability matrix at decision epoch t is:

$$D_t = \begin{bmatrix} p_{1,1} + 0.5\delta_{1,1}(s^*, k^*) & \dots & p_{1,M} + 0.5\delta_{1,M}(s^*, k^*) \\ \vdots & \ddots & \vdots \\ p_{M,1} + 0.5\delta_{M,1}(s^*, k^*) & \dots & p_{M,M} + 0.5\delta_{M,M}(s^*, k^*) \end{bmatrix}.$$

We can see that the transition probability matrix is influenced by the actions, and their intensities, chosen by the decision maker and its competitors. The transition probability matrix can change from epoch to epoch. A sequence of transition matrices over a period of time forms a policy, plan, strategy, or scenario of actions. The n^{th} policy, π_n , is represented as:

$$\pi_n = \{D_{t=1}^n, D_{t=2}^n, \dots, D_{t=N}^n\}. \quad (3)$$

Note that in general, the action transition probabilities in most MDP formulations are constants, and the aim is to find the optimal policy, π^* , that maximizes a certain reward function. In our case, we aim to predict and analyse market shares, and compare possible outcomes of applying different policies.

Market shares of providers at decision epoch t , where $t > 1$, can be calculated as follows:

$$X_2 = X_1 D_1,$$

$$X_t = X_1 \prod_{m=1}^{t-1} D_m, \quad (4)$$

$$X_t = [x_t(s_1) \quad x_t(s_2) \quad x_t(s_3) \quad \dots \quad x_t(M)].$$

The sequence of state vectors is a Markov chain (MC):

$$X = \{X_1, X_2, X_3, \dots\}.$$

As the actions applied are time-varying, so too are the transition matrices D_t . We can view the rightmost term in (2) as perturbations injected into a time-homogeneous MC with a constant transition matrix P . Consequently, X becomes a time-inhomogeneous MC with time-varying transition matrices D_t .

It is important to assume that our model satisfies the Markov condition, i.e. transition probabilities depend only on the present state and actions, but not earlier ones [7].

The reward matrix, R_j , is defined as the reward for provider j . The entries in $R_j, r_{i,j}$, can be revenue, profit, or other measures in real or relative terms. In general, $r_{i,j}$ represents the reward when customers churn from provider i to j . If provider j tries to win market share at a loss, then $r_{i,j}$ would be a negative value.

The value vector, V_{t+1} , shows the reward reaped by keeping existing customers, and gaining customers from other providers during the time period from decision epoch t to $(t + 1)$. The entries of vector V_{t+1} are given as:

$$v_{t+1}(s_j) = u \sum_{i=1}^M x_t(s_i) (d_{t|i,j} \times r_{i,j}) \quad (5)$$

The accumulated value vector, i.e. the accumulated reward reaped over the time period from the beginning ($t = 1$) to decision epoch $t = N$, is given as:

$$W_N = \sum_{t=2}^N V_t \quad (6)$$

Each entry in the vector, $w_t(s_j)$, is the total reward contributed by the customer base of provider j , accumulated from the beginning to $t = N$. Two numerical examples are given in the next section to illustrate how this model can be applied in market share analysis.

IV. NUMERICAL EXAMPLES

A. Example 1 – Market Share, Revenue, and Profit Forecast

There are three major providers and a number of minor providers in a mature mobile market. These providers make decisions on a monthly basis according to their business strategies and market condition. This example illustrates how a provider can make use of the MDP model to predict its market share and revenue over a six month period, under two different scenarios.

Decision epochs are monthly, $T = \{1, 2, \dots, 7\}$. The provider space is $S = \{s_1, s_2, \dots, s_5\} = \{1, 2, \dots, 5\} = \{A, B, C, S, F\}$. Symbols, numbers, and capital letters are used interchangeably in our discussion. Provider A denotes the largest incumbent telephone company, B the second largest, and C the third largest; while S denotes the aggregate of all the smaller providers, and F the future market for growth. The total user population, u , is 30 million including F. The initial distribution vector of system states is given as:

$$X_1 = [x_1(A) \quad x_1(B) \quad x_1(C) \quad x_1(S) \quad \dots \quad x_1(F)],$$

$$X_1 = [0.50 \quad 0.30 \quad 0.15 \quad 0.03 \quad 0.02].$$

These numbers are typical for a mature mobile market, with a market penetration rate well above 100% of the country's population of 22 million. The potential market growth is therefore very small, e.g., $F=0.02$. From the perspective of A, the largest provider, our main interest is to defend from losing market shares to provider B and C. To simplify our analysis, we assume that $x_t(S)$ and $x_t(F)$ will remain constant. Based on past history, we have the following BAU transition probability matrix:

$$P = \begin{bmatrix} 0.9250 & 0.0500 & 0.0250 & 0 & 0 \\ 0.0030 & 0.9850 & 0.0120 & 0 & 0 \\ 0.0750 & 0.0300 & 0.8950 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

From the first row of P , provider A has a retention rate of 92.5%, and loses 5% and 2.5% of its market share to provider B, and C, respectively every month. Provider B has the best retention rate of 98.5%, due to high service quality and slightly better value than provider A. However, provider B loses 1.2% of its market share to provider C due to lower rates offered by C. Although provider C is the cheapest, it has the lowest retention rate of 89.5% due to unstable service and poor brand image. Provider C loses 7.5% of its market share to A every month.

As A is the incumbent operator, it owns the public switched telephone network (PSTN) infrastructure and is in a better position to offer mobile services bundled with other communication services such as broadband and fixed line. As

provider A has already captured the largest market share, A focuses on marketing and offering service bundles, rather than price reduction, as strategies to maintain its market position. Provider B and C are mobile operators and do not own any PSTN infrastructure, they rely on the use of marketing and price reduction strategies to gain market share. Suppose A_A is the action space for provider A, and there are two allowable actions for A, $A_A = \{a_{A,1}, a_{A,2}\}$, and

- $a_{A,1}$: Intense marketing.
- $a_{A,2}$: New service plans, e.g. service bundles.

For provider B, the action space is $A_B = \{a_{B,1}, a_{B,2}\}$, and

- $a_{B,1}$: Intense marketing.
- $a_{B,2}$: Aggressive price reduction.

For provider C, the action space is $A_C = \{a_{C,1}, a_{C,2}\}$, and

- $a_{C,1}$: Intense marketing.
- $a_{C,2}$: Aggressive price reduction.

A provider sets up the delta matrices based on past data, industry knowledge, and experience. Each action has a corresponding delta matrix. For example, the elements of $\Delta(A, 2)$ represent the probabilities added to the BAU matrix when provider A takes action 2, from its own set of allowable actions, at full or 100% intensity. There are six delta matrices in total, two per provider. The matrices $\Delta(A, 1)$, $\Delta(A, 2)$, and $\Delta(C, 2)$ are given below:

$$\Delta(A, 1) = \begin{bmatrix} 0.0300 & -0.0200 & -0.0100 & 0 & 0 \\ 0.0220 & -0.0200 & -0.0020 & 0 & 0 \\ 0.0400 & -0.0050 & -0.0350 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix};$$

$$\Delta(A, 2) = \begin{bmatrix} 0.0380 & -0.0200 & -0.0180 & 0 & 0 \\ 0.0110 & -0.0100 & -0.0010 & 0 & 0 \\ 0.0010 & -0.0001 & -0.0009 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

$$\Delta(C, 2) = \begin{bmatrix} -0.0050 & -0.0100 & 0.0150 & 0 & 0 \\ -0.0010 & -0.0120 & 0.0130 & 0 & 0 \\ -0.0200 & -0.0100 & 0.0300 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The matrices $\Delta(B, 1)$, $\Delta(B, 2)$, and $\Delta(C, 2)$ are not required as the corresponding actions are not taken in this example. For simplicity from now on we will omit the last two columns and rows in the calculations.

From the perspective of provider A, we will look at two scenarios, over a period of six months, i.e. from the beginning of month 1 ($t = 1$), to the end of month 6 ($t = 7$). These two scenarios are based on industry rumors that provider C will start to reduce rates aggressively next month in order to gain market share. In scenario 1, it will be business as usual (BAU) at $t = 1$. Provider C will then take action $a_{C,2}$ at full intensity from $t = 2$ onward. All other providers will not take any action for the whole period of time.

As provider A, we want to see if we can defend our market share through additional marketing and the launching of new service bundles. In scenario 2, in addition to all the actions

taken in scenario 1, provider A will take action $a_{A,1}$ at 30% intensity from $t = 2$ onward and action $a_{A,2}$ at 100% intensity from $t = 3$ onward.

The scenarios are shown as $\pi_1 = \{D_1^1, D_2^1, \dots, D_6^1\}$ and $\pi_2 = \{D_1^2, D_2^2, \dots, D_6^2\}$. The scaling factors of the action matrices for scenario π_1 are:

$$L_1^1(A) = L_2^1(A) = \dots = L_6^1(A) = [0 \ 0],$$

$$L_1^1(B) = L_2^1(B) = \dots = L_6^1(B) = [0 \ 0], \text{ and}$$

$$L_1^1(C) = [0 \ 0]; L_2^1(C) = \dots = L_6^1(C) = [0 \ 1].$$

The scaling factors for scenario π_2 are:

$$L_1^2(A) = [0 \ 0]; L_2^2(A) = [0.3 \ 0];$$

$$L_3^2(A) = \dots = L_6^2(A) = [0.3 \ 1],$$

$$L_1^2(B) = L_2^2(B) = \dots = L_6^2(B) = [0 \ 0], \text{ and}$$

$$L_1^2(C) = [0 \ 0]; L_2^2(C) = \dots = L_6^2(C) = [0 \ 1].$$

From Eqn. 1, for scenario 1,

$$D_1^1 = P = \begin{bmatrix} 0.925 & 0.050 & 0.025 \\ 0.003 & 0.985 & 0.012 \\ 0.075 & 0.030 & 0.895 \end{bmatrix};$$

$$D_2^1 = \dots = D_6^1 = P + 1 \times \Delta(C, 2)$$

$$= \begin{bmatrix} 0.925 & 0.050 & 0.025 \\ 0.003 & 0.985 & 0.012 \\ 0.075 & 0.030 & 0.895 \end{bmatrix} + \begin{bmatrix} -0.005 & -0.010 & 0.015 \\ -0.001 & -0.012 & 0.013 \\ -0.020 & -0.010 & 0.030 \end{bmatrix}$$

$$= \begin{bmatrix} 0.920 & 0.040 & 0.040 \\ 0.002 & 0.973 & 0.025 \\ 0.055 & 0.020 & 0.925 \end{bmatrix}.$$

For scenario 2,

$$D_1^2 = P = \begin{bmatrix} 0.925 & 0.050 & 0.025 \\ 0.003 & 0.985 & 0.012 \\ 0.075 & 0.030 & 0.895 \end{bmatrix};$$

$$D_2^2 = P + 0.3\Delta(A, 1) + \Delta(V, 2) = \begin{bmatrix} 0.9290 & 0.0340 & 0.0370 \\ 0.0086 & 0.9670 & 0.0244 \\ 0.0670 & 0.0185 & 0.9145 \end{bmatrix};$$

$$D_3^2 = \dots = D_6^2 = P + 0.3\Delta(A, 1) + \Delta(A, 2) + \Delta(C, 2)$$

$$= \begin{bmatrix} 0.9670 & 0.0140 & 0.0190 \\ 0.0196 & 0.9570 & 0.0234 \\ 0.0680 & 0.0184 & 0.9136 \end{bmatrix}.$$

Now we have all the transition matrices for both scenarios. We are ready to determine the market shares at each decision epoch. For example, the space vector for scenario 1 at $t = 7$,

$$X_7^1 = X_1 D_1^1 D_2^1 D_3^1 D_4^1 D_5^1 D_6^1 = [0.3581 \ 0.3802 \ 0.2117].$$

TABLE I. SUMMARY OF MARKET SHARES

Epoch (t)	Scenario 1 (π_1)			Scenario 2 (π_2)		
	A	B	C	A	B	C
1	0.5000	0.3000	0.1500	0.5000	0.3000	0.1500
2	0.4747	0.3250	0.1504	0.4747	0.3250	0.1504
3	0.4456	0.3382	0.1662	0.4538	0.3332	0.1630
4	0.4198	0.3502	0.1800	0.4565	0.3282	0.1653
5	0.3968	0.3612	0.1920	0.4591	0.3235	0.1674
6	0.3763	0.3711	0.2025	0.4616	0.3191	0.1692
7	0.3581	0.3802	0.2117	0.4642	0.3150	0.1708

The market shares of the two scenarios are summarized in Table 1. The market share is based on $u = 30$ million subscriptions in total.

The first two rows in the table show the current trend under the BAU condition. Provider A is mainly losing its market share to provider B; whilst provider C is maintaining its position with slight increase. Under scenario 1, if provider A does not take any counter action, it will see a significant drop in market share, from 50% to 35.81%. Under scenario 2, where provider A takes actions in extra marketing and offering service bundles, its market share goes back up steadily. Provider C continues to gain market share but at a slower pace. Provider B begins to lose its market share gradually.

The next step is to calculate the financial rewards for provider A. Suppose provider A

- earns a revenue of \$25 per month per subscriber,
- makes profit of \$5 per month per existing subscriber,
- incurs a once off acquisition cost of \$4 per new subscriber.

The reward matrices for revenue and profit calculations are given as $R_A^{revenue}$, and R_A^{profit} respectively:

$$R_A^{revenue} = \begin{bmatrix} 25 & 0 & 0 \\ 25 & 0 & 0 \\ 25 & 0 & 0 \end{bmatrix}, R_A^{profit} = \begin{bmatrix} 5 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

The following shows how we calculate the revenue value vector V_2^1 , following scenario 1, π_1 , at decision epoch $t = 2$.

$$V_2^1 = 30 \times 10^6 \begin{bmatrix} 0.5 \\ 0.3 \\ 0.15 \end{bmatrix}^T \begin{bmatrix} 0.925 \times 25 & 0.050 \times 0 & 0.025 \times 0 \\ 0.003 \times 25 & 0.985 \times 0 & 0.012 \times 0 \\ 0.075 \times 25 & 0.030 \times 0 & 0.895 \times 0 \end{bmatrix} \\ = [338,188,125 \ 0 \ 0].$$

Provider A achieves revenue of over 338 million dollars in the first month. The second and third entries in V_2^1 show the obvious, that other providers retaining or gaining subscribers contribute nothing to provider A's revenue.

A full summary of results is given in Table 2 shown on the next page.

Based on the numbers above, the actions taken by provider A, as specified in π_2 , are effective to defend its market share and profit margin over the period of six months, if provider C takes actions according to π_1 .

TABLE II. SUMMARY OF REVENUES AND PROFIT FOR PROVIDER A (IN MILLION DOLLARS)

Epoch (t)	Scenario 1 (π_1)		Scenario 2 (π_2)	
	Revenue	Profit	Revenue	Profit
2	338.19	66.25	338.19	66.25
3	317.49	62.48	323.35	63.20
4	299.08	58.70	325.22	63.04
5	282.71	55.33	327.09	63.40
6	268.13	52.34	328.92	63.76
7	255.15	49.68	330.72	64.12
Total:	1760.75	344.78	1973.49	383.78

B. Example 2 – Merging Two Providers

Although provider A holds the largest market share, its market position is under threat and is gradually losing market share to other players. To ensure its dominance, provider A acquires provider C to become the dominant player, now called AC. Assume that after acquisition, both providers A and C will continue to run independently under the umbrella of AC.

Suppose the scenarios π_1 and π_2 described in Example 1 are followed. This example shows how revenue and profit can be determined by this newly created market leader.

As provider C is a low cost provider, targeting the lower end of the market, its revenue and profit margin are relatively lower. Suppose provider C

- earns a revenue of \$18 per month per subscriber,
- makes profit of \$3 per month per existing subscriber,
- incurs a once off acquisition cost of \$4.50 per new subscriber.

The reward matrices for provider AC are:

$$R_{AC}^{revenue} = R_A^{revenue} + R_C^{revenue}; \text{ and}$$

$$R_{AC}^{profit} = R_A^{profit} + R_C^{profit}.$$

Therefore,

$$R_{AC}^{revenue} = \begin{bmatrix} 25 & 0 & 18 \\ 25 & 0 & 18 \\ 25 & 0 & 18 \end{bmatrix}; R_{AC}^{profit} = \begin{bmatrix} 5 & 0 & (3 - 4.5) = -1.5 \\ 1 & 0 & (3 - 4.5) = -1.5 \\ 1 & 0 & 3 \end{bmatrix}.$$

Following scenario 1, π_1 , at decision epoch $t = 2$, the value vector is:

$$V_2^1 = 30 \times 10^6 \begin{bmatrix} 0.5 \\ 0.3 \\ 0.15 \end{bmatrix}^T \begin{bmatrix} 0.925 \times 25 & 0.050 \times 0 & 0.025 \times 18 \\ 0.003 \times 25 & 0.985 \times 0 & 0.012 \times 18 \\ 0.075 \times 25 & 0.030 \times 0 & 0.895 \times 18 \end{bmatrix}$$

$$= [338,188,125 \quad 0 \quad 77,129,550].$$

In the first month provider AC receives total revenue of over 415 million dollars, of which approximately 338 million dollars are contributed by provider A, 77 million dollars by provider C. Revenues and profits are summarized in Table 3 and Table 4, respectively. Based on our modelling, scenario 2 provides a better outlook for both revenue and profit in the coming six months.

TABLE III. SUMMARY OF REVENUES FOR EXAMPLE 2 (IN MILLION DOLLARS)

Epoch (t)	Scenario 1 (π_1)			Scenario 2 (π_2)		
	Revenue by A	Revenue by C	Total	Revenue by A	Revenue by C	Total
2	338.19	77.13	415.32	338.19	77.13	415.32
3	317.49	85.25	402.74	323.35	83.61	406.96
4	299.08	92.34	391.42	325.22	84.81	410.04
5	282.71	98.52	381.23	327.09	85.87	412.96
6	268.13	10.39	372.04	328.92	86.81	415.73
7	255.15	10.86	363.74	330.72	87.64	418.36
Total:	1760.75	565.74	2326.49	1973.49	505.88	2479.37

TABLE IV. SUMMARY OF PROFITS FOR EXAMPLE 2 (IN MILLION DOLLARS)

Epoch (t)	Scenario 1 (π_1)			Scenario 2 (π_2)		
	Profit by A	Profit by C	Total	Profit by A	Profit by C	Total
2	66.25	10.79	77.04	66.25	10.79	77.04
3	62.48	10.73	73.21	63.20	10.67	73.87
4	58.70	12.02	70.72	63.04	12.03	75.07
5	55.33	13.14	68.48	63.40	12.21	75.62
6	52.34	14.12	66.46	63.76	12.38	76.14
7	49.68	14.98	64.65	64.12	12.52	76.64
Total:	344.78	75.79	420.57	383.78	70.60	454.38

V. CONCLUSION

We presented a finite-horizon, discrete time MDP model for mobile service providers to model decisions and forecast market shares and financial rewards. The model and examples demonstrate how MDPs offer a framework to evaluate business strategies for the telecommunications industry.

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